



Book Introduction: Traffic Flow Dynamics Data, Models and Simulation

Chapter.9: Macroscopic Models with Dynamic Velocity



Macroscopic Acceleration Function

The Limitation of first order model

- In first-order (Lighthill-Whitham-Richards, LWR) models, the local speed and flow are **statically coupled** to the **density** by the fundamental relation. This implies instantaneous adaptation to new circumstances and leads to unbounded accelerations and other unrealistic consequences such as the **lack of hysteresis effects** or **traffic instabilities**.
- First-order models are characterized by a single dynamical partial differential equation (or iterated coupled map) for the density
- From a microscopic point of view, the associated instantaneous speed adaptations imply unbounded accelerations (unrealistic)
- Finite speed adaptation times and reaction times are the main factors leading to growing traffic waves and capacity-drop phenomena , or to traffic flow instabilities in general .

Macroscopic Acceleration Function

Second order model

- The local speed possesses its own dynamical acceleration equation describing speed changes as a function of density, local speed, their gradients, and possibly other exogenous factors.
- Second-order models are the models of choice to macroscopically describe traffic-flow instabilities leading to traffic waves, the capacity drop phenomenon, or scattered flow-density data.
- The local speed is treated as a second independent field which is governed by a second dynamical acceleration equation. Such an equation describes the local acceleration as a function of density, speed, gradients thereof, and possibly other exogenous factors.
- This class of models is also known as the class of second-order models, in contrast to the LWR models which are also termed first-order models.

Macroscopic Acceleration Function

Second-order model

- In time-continuous second-order models,² the acceleration equation is a second partial differential equation of the general form

$$\frac{dV(x, t)}{dt} = \left(\frac{\partial}{\partial t} + V(x, t) \frac{\partial}{\partial x} \right) V(x, t) = A[\rho(x, t), V(x, t)]. \quad (9.1)$$

Total time derivative (left-hand side) consists of two parts: **the local rate of speed change** $\frac{\partial V}{\partial t}$ and the **convective rate of change** $\frac{\partial V}{\partial x}$ due to moving to a new location

Spatial anticipation

- The different second-order models are solely distinguished by their acceleration function, characterizing the macroscopic acceleration as a function of the density and speed field of the neighborhood.
- However, just a function is not sufficient since this would correspond to very short-sighted drivers not taking into account what happens in front of them.

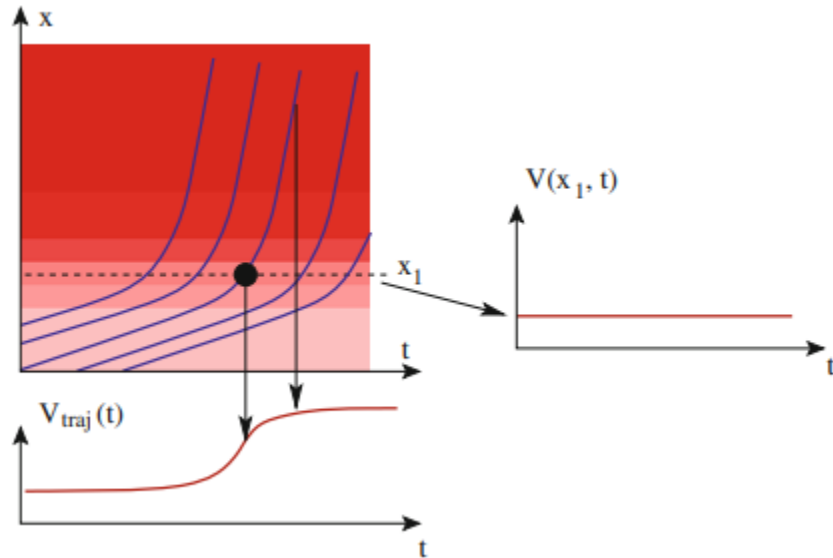
Macroscopic Acceleration Function

Spatial anticipation

- There are 2 methods to introduce Spatial anticipation:
 1. Allowing density and speed gradients $\frac{\partial V}{\partial t}, \frac{\partial \rho}{\partial t}$ (and possibly higher-order gradients)
 2. Take the density and speed fields at the spatial coordinate $x_a > x$ in front of the actual position x

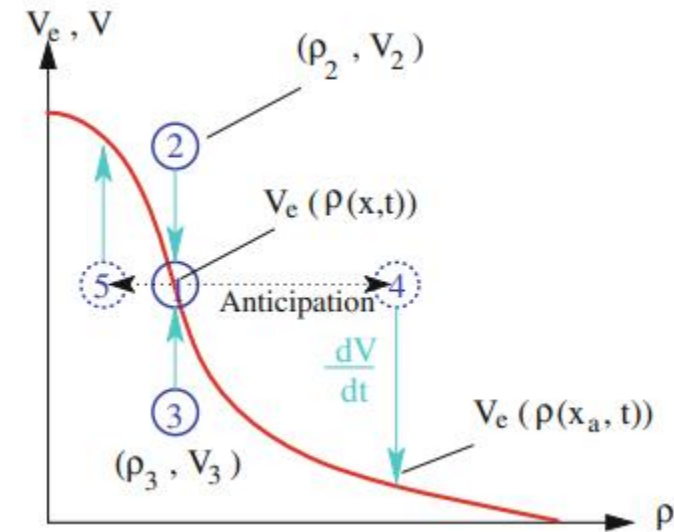
$$\rho_a(x, t) = \rho(x_a, t), \quad V_a(x, t) = V(x_a, t), \quad x_a > x. \quad (9.2)$$

Macroscopic Acceleration Function



Along one trajectory,

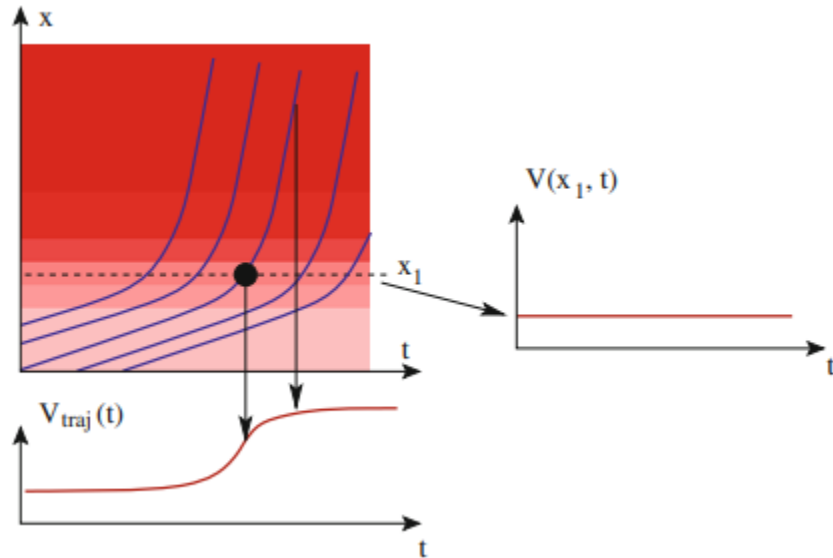
$$\frac{dV(x(t), t)}{dt} = \frac{\partial V}{\partial x} \frac{dx}{dt} + \frac{\partial V}{\partial t} = V \frac{\partial V}{\partial x} + \frac{\partial V}{\partial t}$$



1 to 3 characterize local situations (ρ, V) in steady state (1) and in non-equilibrium (2 and 3)

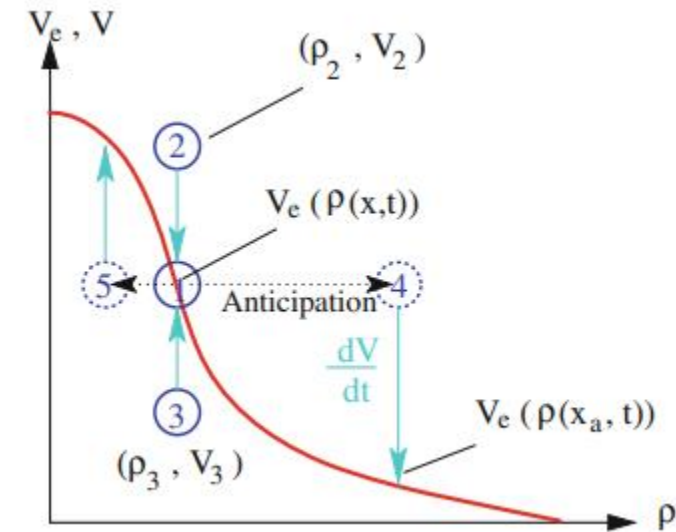
The arrows from the states 2 and 3 to 1 describe local speed adaptation, while the arrows from 4 and 5 describe anticipative adaptation.

Macroscopic Acceleration Function



Along one trajectory,

$$\frac{dV(x(t), t)}{dt} = \frac{\partial V}{\partial x} \frac{dx}{dt} + \frac{\partial V}{\partial t} = V \frac{\partial V}{\partial x} + \frac{\partial V}{\partial t}$$



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The arrows from the states 2 and 3 to 1 describe local speed adaptation, while the arrows from 4 and 5 describe anticipative adaptation.

Macroscopic Acceleration Function

Since both gradients and nonlocalities describe the same aspect (spatial anticipation), they are interchangeable. Therefore, local macroscopic models can be written in the general form.

$$\frac{dV}{dt} = \left(\frac{\partial}{\partial t} + V \frac{\partial}{\partial x} \right) V = A_{\text{loc}} \left(\rho, V, \frac{\partial \rho}{\partial x}, \frac{\partial V}{\partial x} \right), \quad (9.3)$$

While nonlocal second-order macroscopic models take on the general form

$$\frac{dV}{dt} = \left(\frac{\partial}{\partial t} + V \frac{\partial}{\partial x} \right) V = A_{\text{nonloc}} (\rho, V, \rho_a, V_a). \quad (9.4)$$

Computationally faster and numerically more stable than local models

Properties of the Acceleration Function

Steady-State Flow

Second-order models possess a unique steady state on homogeneous roads described by the steady-state speed $V_e(\rho)$ for a given density. With the hydrodynamic relation $Q = \rho V$, this results in a fundamental diagram $Q_e(\rho)$

For stationary-homogeneous conditions, acceleration functions are equal to 0.

Considering local and nonlocal acceleration functions for local & nonlocal second-order, respectively, and taking into account stationarity $\frac{\partial}{\partial t} = 0$ and homogeneity $\frac{\partial}{\partial x} = 0$, $X_a = x$, we arrive at the steady-state conditions

$$A_{\text{loc}}(\rho, V_e, 0, 0) = 0, \quad A_{\text{nonloc}}(\rho, V_e, \rho, V_e) = 0.$$

The conditions for plausible speed-density relations are the same as for the LWR models

$$\left\{ \begin{array}{l} V'_e(\rho) \leq 0 \\ V_e(0) = V_0 \\ V_e(\rho_{\max}) = 0 \end{array} \right.$$

Properties of the Acceleration Function

Plausibility Conditions

In contrast to the LWR models, the local speed $V(x, t)$ is generally not equal to the steady-state speed $V_e(\rho(x, t))$. Nevertheless, the acceleration function should model the desire of the drivers to approach the steady state. In homogeneous situations (no speed or density gradients) this leads to following criteria for plausible acceleration functions, i.e., plausible models:

$$A(\rho, V, \dots) < 0, \quad \text{if } V > V_e(\rho), \quad \frac{\partial \rho(x, t)}{\partial x} = 0, \quad \text{and} \quad \frac{\partial V(x, t)}{\partial x} = 0. \quad (9.6)$$

This condition states that, when increasing the local speed and leaving everything else unchanged (*ceteris paribus*), the acceleration decreases.

Analogously, $A(\rho, V, \dots) > 0$ if $V < V_e(\rho)$. We can summarize both conditions by demanding that the partial derivative of the acceleration functions with respect to the local speed should be strictly negative:

$$\frac{\partial A_{\text{loc}}\left(\rho, V, \frac{\partial \rho}{\partial x}, \frac{\partial V}{\partial x}\right)}{\partial V} < 0, \quad \frac{\partial A_{\text{nonloc}}(\rho, V, \rho_a, V_a)}{\partial V} < 0. \quad (9.7)$$

Properties of the Acceleration Function

Plausibility Conditions

- The existence of a steady-state, and drivers attempting to reach it is not sufficient since this alone would result in unconditionally unstable traffic flow:
- Models without additional negative feedback mechanisms would always produce traffic waves, even in completely free traffic, which is even more unrealistic than the total absence of such waves in the LWR models.
- **In real traffic, instabilities are only observed in a density range corresponding to congested traffic.**
- This leads us to the third set of plausibility conditions:

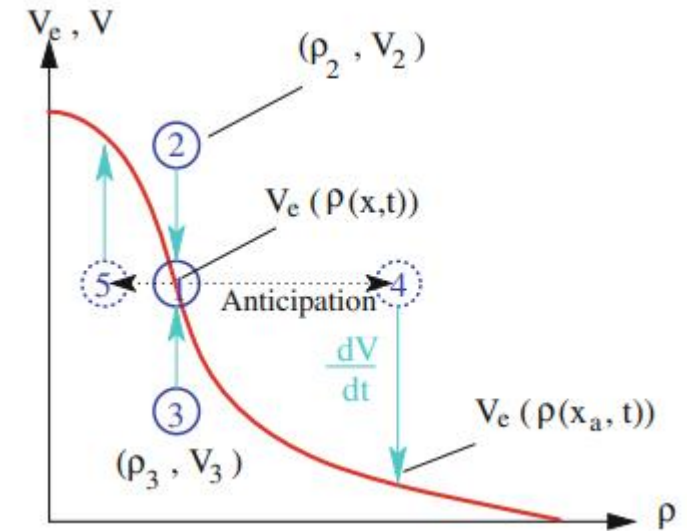
$$\frac{\partial A \left(\rho, V, \frac{\partial \rho}{\partial x}, \frac{\partial V}{\partial x} \right)}{\partial \left(\frac{\partial \rho}{\partial x} \right)} \leq 0, \quad \frac{\partial A \left(\rho, V, \frac{\partial \rho}{\partial x}, \frac{\partial V}{\partial x} \right)}{\partial \left(\frac{\partial V}{\partial x} \right)} \geq 0. \quad (9.8)$$

Properties of the Acceleration Function

Plausibility Conditions

When modeling anticipation by nonlocalities, a plausible driver's behavior is even more explicit :Reduce speed if the density ρ_a ahead is higher than the local density ρ , increase speed if the speed V_a ahead is higher than the local speed V . Moreover, the degree of reaction should increase monotonically with the differences, so we require

$$\frac{\partial A(\rho, V, \rho_a, V_a)}{\partial \rho_a} \leq 0, \quad \frac{\partial A(\rho, V, \rho_a, V_a)}{\partial V_a} \geq 0. \quad (9.9)$$



General Form of the Model Equations

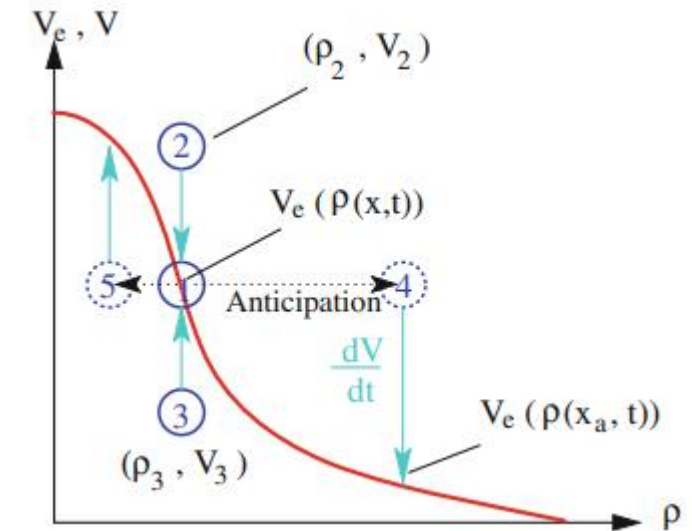
- We are going to formulate general forms for effectively **single-lane** and **single-class second-order** macroscopic models. Since vehicle conservation is always valid, second-order models obey the same continuity equations as LWR models, in the most general case with ramps and changes of the number of lanes.
- Acceleration equation may also contain diffusion terms—second-order spatial derivatives.
- In order to arrive at a compact formulation, we express the speed and density gradients in terms of a gradient of a traffic pressure P while the nonlocalities can be expressed in terms of a generalized targeted speed $V_e^*(\rho, V, \rho_a, V_a)$ that can be seen as a generalization of the steady-state speed $V_e^*(\rho)$.
- Finally, on road sections parallel to the acceleration and deceleration lanes of ramps, the acceleration equation generally contains additional ramp terms describing the influence of entering or exiting vehicles on the local speed of the main road

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho V)}{\partial x} = \frac{\partial}{\partial x} \left(D \frac{\partial \rho}{\partial x} \right) - \frac{\rho V}{I} \frac{dI}{dx} + v_{\text{rmp}}(x, t) \quad (9.10)$$

$$\boxed{\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} = \frac{V_e^*(\rho, V, \rho_a, V_a) - V}{\tau} - \frac{1}{\rho} \frac{\partial P}{\partial x} + \frac{1}{\rho} \frac{\partial}{\partial x} \left(\eta \frac{\partial V}{\partial x} \right) + A_{\text{rmp}}(x, t).} \quad (9.11)$$

General Form of the Model Equations

- Local speed adaption
- In local models, the generalized targeted speed V_e^* is equal to the local steady-state speed $V_e(\rho)$ and the first term on the right-hand side of function, also denoted as speed adaptation term or relaxation term describes the mean acceleration of the vehicles in the local neighborhood in order to reach the steady-state speed corresponding to the local density.
- the speed adaptation time is constant (does not depend on density or speed), and the acceleration is proportional to the difference of the local speed V to the steady-state speed $V_e(\rho)$
- Then, τ represents a characteristic speed adaptation time in which the distance to the steady-state is $1/e$ times the original distance.



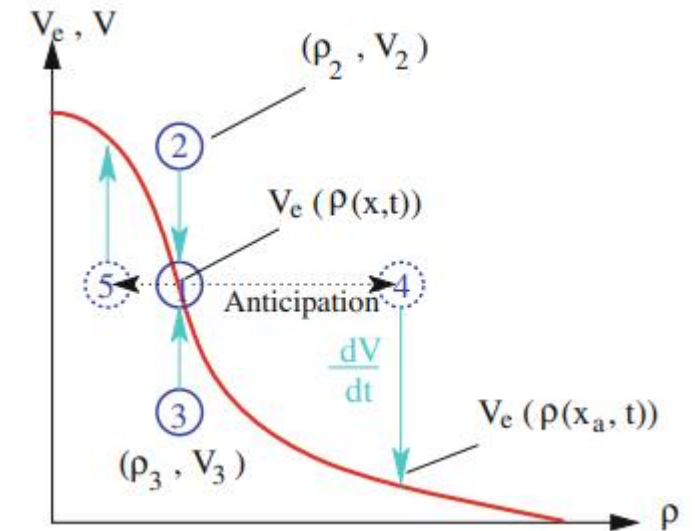
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General Form of the Model Equations

- Nonlocal Anticipation
- In nonlocal models, the generalized targeted speed V_e^* of the speed adaptation term depends not only on the local density but on the density $\rho_a = \rho(x_a, t)$ at an anticipated location ahead of the actual position, $x_a > x$, and possibly on the actual and anticipated speeds V and V_a , respectively. In the simplest case, $V_e^* = V_e(\rho_a)$ is directly given by the steady-state speed at the anticipated location resulting in the nonlocal adaptation term.

$$A_{\text{relax+antic}}(x, t) = \frac{V_e(\rho(x_a, t)) - V(x, t)}{\tau}. \quad (9.12)$$

Thus, an anticipatory driving style is simply described by adapting the speed to the steady-state speed as in local models, but taking the steady-state speed at a position ahead.



General Form of the Model Equations

Pressure Term

The pressure term introducing the traffic pressure P describes a response of the local ensemble of vehicles or drivers on density gradients and, in some models, speed gradients.

$$-\frac{1}{\rho} \frac{\partial P}{\partial x} = -\frac{1}{\rho} \frac{\partial P_{\text{antic}}(\rho, V)}{\partial x} - \frac{1}{\rho} \frac{\partial P_{\text{kin}}(\rho, V)}{\partial x} \quad (9.13)$$

This term describes a purely kinematic (statistical) effect of speed variance without a single vehicle accelerating or braking. Besides this first-principles interpretation, traffic pressure terms are also used purely phenomenologically and then, indeed, describe the anticipation of the drivers. Therefore, the pressure term generally has both a behavioral and a kinematic component

$$A_{\text{antic}} = -\beta_1 \frac{\partial \rho}{\partial x} + \beta_2 \frac{\partial V}{\partial x}, \quad \beta_1 = \frac{1}{\rho} \frac{\partial P_{\text{antic}}}{\partial \rho}, \quad \beta_2 = -\frac{1}{\rho} \frac{\partial P_{\text{antic}}}{\partial V} \quad (9.14)$$

$$\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} = \frac{V_e^*(\rho, V, \rho_a, V_a) - V}{\tau} - \frac{1}{\rho} \frac{\partial P}{\partial x} + \frac{1}{\rho} \frac{\partial}{\partial x} \left(\eta \frac{\partial V}{\partial x} \right) + A_{\text{rmp}}(x, t). \quad (9.11)$$

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General Form of the Model Equations

Pressure Term

In contrast to the behavioral part of traffic pressure representing anticipative driver reactions, the kinematic part is a sole consequence of a finite speed variance of the vehicles i in the microscopically large and macroscopically small local neighborhood.

The vehicles contributing to the variance are exactly the same that contribute to the local speed, $V(x, t) = v_i$. The kinematic pressure term is only effective if

- (i) the speed variance is nonzero
- (ii) there are density gradients.

Then, it leads to a macroscopic speed change even if not a single vehicle accelerates.

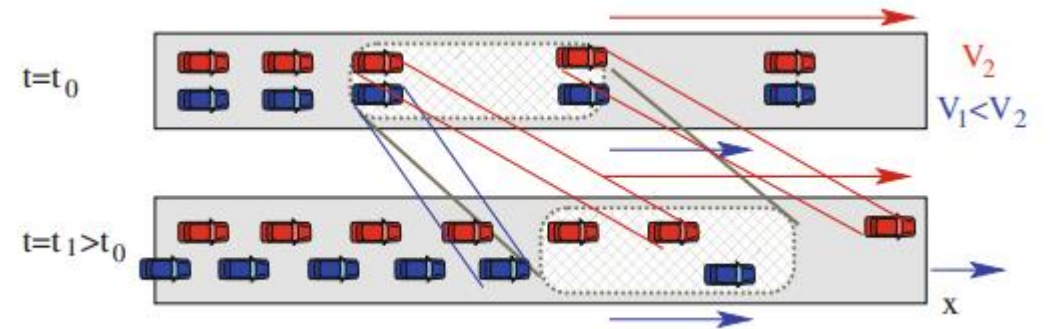
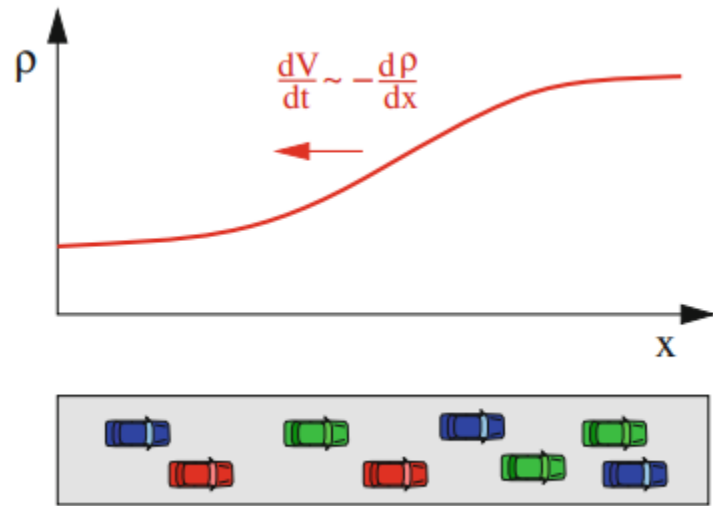
$$-\frac{1}{\rho} \frac{\partial P}{\partial x} = -\frac{1}{\rho} \frac{\partial P_{\text{antic}}(\rho, V)}{\partial x} - \frac{1}{\rho} \frac{\partial P_{\text{kin}}(\rho, V)}{\partial x} \quad (9.13)$$

$$A_{\text{kin}} = -\frac{1}{\rho} \frac{\partial P_{\text{kin}}}{\partial x}, \quad P_{\text{kin}} = \rho \sigma_V^2(x, t), \quad (9.15)$$

$$A_{\text{kin}} = -\frac{1}{\rho} \frac{\partial}{\partial x} \left(\rho(x, t) \sigma_V^2(x, t) \right) \approx -\frac{\sigma_V^2}{\rho} \frac{\partial \rho}{\partial x} > 0$$

$$\boxed{\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} = \frac{V_e^*(\rho, V, \rho_a, V_a) - V}{\tau} - \frac{1}{\rho} \frac{\partial P}{\partial x} + \frac{1}{\rho} \frac{\partial}{\partial x} \left(\eta \frac{\partial V}{\partial x} \right) + A_{\text{rmp}}(x, t).} \quad (9.11)$$

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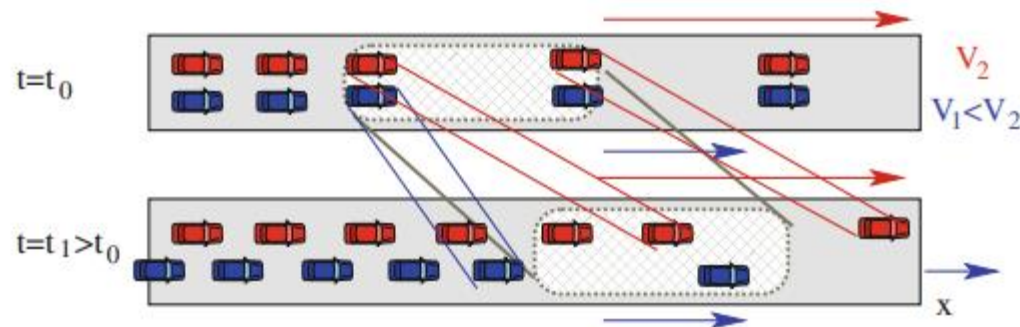
General Form of the Model Equations

On- and Off-Ramp Terms

As is the case for the kinematic pressure term, A_{rmp} describes changes of the macroscopic local speed by changes in the vehicle composition rather than by microscopic accelerations.

Takes into consideration that the right-hand side of the acceleration equation describes the rate of change of the mean speed in microscopically large and macroscopically small road elements comoving with the vehicles (hatched regions)

We calculate this rate of change assuming that on-ramp vehicles merge to the main-road at speed $V_{rmp} < V$. Conversely, we assume that drivers about to leave the main-road reduce their speed, on average, to V_{rmp} before they diverge to the off-ramp. For the case of on-ramps, new vehicles enter the hatched regions at a negative relative speed.



$$A_{rmp}(x, t) = \frac{(V_{rmp} - V)}{\rho} |\nu_{rmp}(x, t)| = \frac{(V_{rmp} - V) |Q_{rmp}|}{\rho I L} \quad (9.17)$$

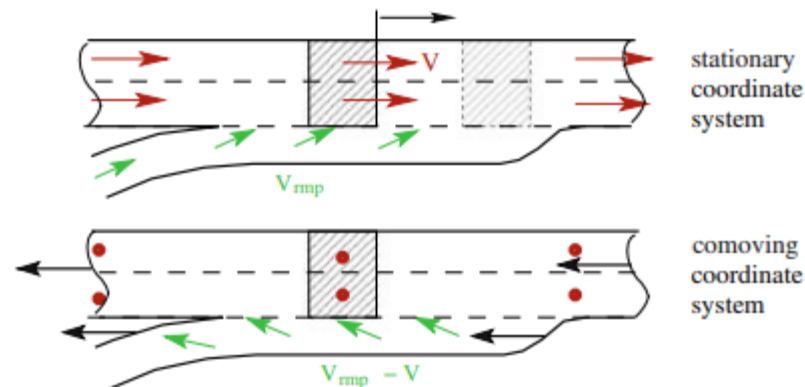
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Overview of Second-Order Models

Payne's Model, 1971

A simple representative of a local second-order macroscopic model is Payne's model, sometimes also called the Payne-Whitham model. Its acceleration equation is

1. constant speed relaxation time τ
2. Zero diffusion η
3. Density dependent traffic pressure
 $P = -Ve(\rho)/2\tau$

$$\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} = \frac{V_e(\rho) - V}{\tau} + \frac{V'_e(\rho)}{2\rho\tau} \frac{\partial \rho}{\partial x} \quad \text{Payne's model.} \quad (9.18)$$

Payne's model is a special case of the general macroscopic acceleration equation

$$\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} = \frac{V_e^*(\rho, V, \rho_a, V_a) - V}{\tau} - \frac{1}{\rho} \frac{\partial P}{\partial x} + \frac{1}{\rho} \frac{\partial}{\partial x} \left(\eta \frac{\partial V}{\partial x} \right) + A_{\text{rmp}}(x, t). \quad (9.11)$$

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Overview of Second-Order Models

Payne's Model, 1971

Traffic pressure. The connection between the traffic pressure and driver interactions can be made more explicit by augmenting the traffic pressure by the constant $V_0/(2\tau)$ resulting in

$$P_{\text{Payne}}(x, t) = \frac{V_0 - V_e(\rho(x, t))}{2\tau}. \quad (9.19)$$

On the other hand, when interpreting the pressure as a purely kinematic effect, we can relate the pressure P_{Payne} to the speed variance according to Eq. (9.15) resulting in

$$\sigma_V^2 = \frac{P_{\text{Payne}}}{\rho} = \frac{V_0 - V_e}{2\tau\rho}.$$

$$A_{\text{kin}} = -\frac{1}{\rho} \frac{\partial P_{\text{kin}}}{\partial x}, \quad P_{\text{kin}} = \rho \sigma_V^2(x, t), \quad (9.15)$$

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When formulating the pressure term as

$$-\frac{1}{\rho} \frac{\partial P}{\partial x} = -\frac{1}{\rho} \frac{\partial}{\partial x} \left(\frac{V_0 - V_e(\rho(x, t))}{2\tau} \right) = -\frac{1}{\rho} \frac{V_e'(\rho)}{2\tau} \frac{\partial \rho}{\partial x} = \frac{c_0^2(\rho)}{\rho} \frac{\partial \rho}{\partial x},$$

Payne's model is formally identical to the equations describing a compressible one-dimensional gas with generally density dependent sonic velocities $\pm c_0 = \pm \sqrt{V_e'/2\tau}$.¹²

Overview of Second-Order Models

Payne's Model, 1971

Parameters and simulation

The only additional parameter is the speed relaxation time τ

- From the microscopic derivation it follows that τ is identical to the reaction time T_r
- While, in reality, these characteristic times have different orders of magnitude: The speed adaptation time is of the order of 10s (more in highway traffic, less in city traffic) while the reaction time is about 1s.
- Typical values for τ adopted in the simulation are between 1 and 5 s. This corresponds to sonic velocities of the order of $\pm 10 \text{ m/s}^2$.

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Payne's model is difficult to simulate and prone to physical and numerical instabilities . Dedicated numerical methods are necessary.

Overview of Second-Order Models

Payne's Model, 1971

Limiting case of the adaptation time tending to zero.

Since the speed relaxation time enters not only the relaxation term of Payne's model but also the pressure term, we will not simply obtain $V = V_e(\rho)$ when multiplying the acceleration equation by τ and setting $\tau = 0$.

$$\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} = \frac{V_e(\rho) - V}{\tau} + \frac{V'_e(\rho)}{2\rho\tau} \frac{\partial \rho}{\partial x} \quad \text{Payne's model.}$$

$$\tau \left(\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} \right) = V_e(\rho) - V + \frac{V'_e(\rho)}{2\rho} \frac{\partial \rho}{\partial x}$$

$$0 = V_e(\rho) - V + \frac{V'_e(\rho)}{2\rho} \frac{\partial \rho}{\partial x} \implies V = V_e(\rho) + \frac{V'_e(\rho)}{2\rho} \frac{\partial \rho}{\partial x}$$